# Finite-Radius Centering for Monodromy Cloud Audits

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**Abstract**

The archived noisy-cloud audit used the limiting monodromy radius, although the exact edge-deflated shell has a finite-period radius. From the RH-17 multiplier law $|M_k| = C_M \lambda^k(1 + o(1))$, we prove that comparing the $2(k-1)$ shell roots with the limiting radius creates an accumulated root-$\ell^1$ bias tending to $\beta|\log C_M|$. In contrast, the radial bias of every fixed pre-alias moment is $O(k^{-1})$. Re-centering the seven RH-15 clouds at archived multiprecision approximations to the finite-cycle radii reduces both total root errors and maximum moment defects. These are floating diagnostics, not interval or asymptotic noisy-cloud transport. The distinction explains why root-$\ell^1$ convergence to the limiting shell is stronger than the control needed for each fixed coefficientwise bridge; uniform determinant gluing still requires a growing weighted-prefix estimate.

## 1 Finite shell radius

Let

$$\beta = \frac{1}{r_H\sqrt{\lambda}}, \qquad \beta_k = \frac{|M_k|^{-1/(2k)}}{r_H}, \qquad r_H = 0.85.$$

RH-17 proves $|M_k| = C_M \lambda^k(1 + o(1))$ for a constant $C_M > 0$. Hence

$$\beta_k = \beta \exp\left[-\frac{\log C_M}{2k} + o(k^{-1})\right]. \tag{1}$$

**Theorem 1.1** (two centering scales)**.** *The radial matching cost between the finite and limiting edge-deflated shells satisfies*

$$2(k-1)|\beta_k - \beta| \longrightarrow \beta|\log C_M|.$$

*For every fixed even $n$,*

$$2|\beta_k^n - \beta^n| = \frac{n\beta^n|\log C_M|}{k} + o(k^{-1}).$$

*Odd shell moments vanish at both radii before the first alias.*

*Proof.* Expand the exponential in (1). There are $2(k-1)$ roots, all receiving the same radial displacement. For the moment statement, raise (1) to the fixed power $n$ and use the exact pre-alias ledger $-2\beta_k^n$ at even orders and zero at odd orders. □

If $C_M \neq 1$, the first limit is positive. The archived multiprecision diagnostic gives $C_M = 1.9463429052\ldots$, but it is not a directed-rounding certificate; positivity of the numerical constant is therefore not promoted to a new computer-assisted theorem here.

## 2 Seven-row re-audit

For an RH-15 cloud of effective degree $N$, we set $k = N + 1$ and use the archived multiprecision approximation to the exact RH-17 radius $\beta_k$. The phase grid remains $j\pi/(N + 1)$, $1 \leq j \leq N$. The same root and moment diagnostics as RH-275 then give:

| quantity | limiting-radius audit | finite-radius audit |
|---|---|---|
| total root error range | 0.6424–1.2481 | 0.2841–0.8992 |
| maximum pre-alias moment range | 0.5096–1.4573 | 0.3640–1.0450 |

All seven finite-radius root errors are smaller than their limiting-radius counterparts. This comparison reuses frozen binary64 roots and multiprecision radii; it is not interval validation.

*Remark* 2.1. The theorem corrects the target of future aggregate audits. It does not select a canonical noisy pole cloud, certify its roots, or prove convergence as $\sigma \downarrow 0$. Triggers 1 and 2 and Gates A–E remain open. No Hilbert–Polya or Riemann-hypothesis conclusion is involved.